

# Abdelrahman Ahmed Al-Meshwady

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**Results-oriented Backend Engineer** with **3+ years of experience specializing in the Node.js ecosystem and IoT/Real-time tracking systems**. Advanced expertise in TypeScript, NestJS, and system design for high-concurrency environments. Proven track record of architecting modular, enterprise-grade systems for Fintech and IoT, combining deep technical execution with a strategic Project Management background to deliver scalable, high-performance applications.

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## Work Experience

**Root Co. Software House | Alexandria, Egypt**

**Backend Engineer | February 2023 - Present**

- Designed and maintained scalable backend services using **Node.js**, leveraging the flexibility of **Express.js** for prototyping and the structured, modular architecture of **NestJS** for enterprise applications and **high-concurrency IoT ecosystems**.
- Engineered RESTful APIs with **TypeScript** to enforce type safety, reduce runtime errors, and improve developer experience through self-documentation.
- Implemented authentication, authorization, and security best practices using JWT, role-based access control, and secure third-party integrations.
- Collaborated with cross-functional teams, participated in code reviews, and supported the deployment and monitoring of backend services in production.

**Project Manager & Technical Lead | April 2022 - January 2024**

- Coordinated cross-functional engineering teams and aligned technical execution with business goals using Agile methodologies.
  - Acted as the main technical point of contact between stakeholders and engineering teams.
  - Oversaw delivery, risk management, and scope alignment for multiple software projects.
  - Led the technical transition of legacy systems to **Node.js**, overseeing the development of real-time modules and IoT integrations.
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## Key Projects

**WeCare App (IoT & Real-time Tracking)**

- Architected the complete backend lifecycle: built the initial production-grade system using Python (Django) for robust business logic, then strategically migrated the real-time tracking module to **Node.js** to scale **WebSocket** concurrency to over **1,000+ users**.
- Designed a containerized microservices architecture using Docker Compose, orchestrating backend services, databases, brokers, and media servers.
- Built real-time bus tracking and proximity alert features using **MQTT** and **Redis geospatial** queries, enabling live trip monitoring.
- Implemented secure, self-hosted live video streaming for bus cameras using **WebRTC** and **HLS**, reducing reliance on third-party services.

**Relive App (Telehealth & RTC)**

- Built a modular, production-ready Node.js backend for Relive, a telehealth platform, with secure multi-role authentication and integrated payments (TAP).
  - Delivered scalable APIs for appointments, prescriptions, content, and notifications, ensuring high performance, maintainability, and robust security.
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## Technical Skills

- Languages & Frameworks: TypeScript, JavaScript (Node.js), NestJS, Express.js, Python, Django, SQL
  - Databases & APIs: MongoDB, PostgreSQL, TypeORM/Prisma, RESTful API Design, JWT Authentication
  - DevOps & Infrastructure: Docker, Nginx, PM2, Ubuntu Server, AWS, Vercel, SSL/DNS Management
  - Tools & Methodologies: Git, GitHub, Agile (Scrum, Kanban), Jira, ClickUp, Trello
  - Collaboration & Design Tools: Figma, Photoshop, Illustrator, Adobe XD
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## Education

- Bachelor of Commerce, Accounting (English Dept.) | Alexandria University, Graduated 2022